WATER SALINITY AND ITS IMPACT IN KALPAR VILLAGE: A PARTICIPATORY RURAL APPRAISAL APPROACH

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ABSTRACT

Groundwater salinity affects 250 million people across 1,20,000-1,50,000 villages in India. In Tamil Nadu's Ramanathapuram district, 90% of groundwater samples exceed WHO standards, forcing coastal communities like Kalpar village, where 100% of wells are non-potable, into economic distress. While hydrogeological studies document salinity levels, household socio-economic impacts remain unquantified. This study adopts participatory methods to investigate: (1) regressive household economic burdens and livelihood impact due to salinity, (2) how PRA-HCD integration reveals community priorities, and (3) socio-ecological feedback loops connecting environmental and social systems. Using Socio-Ecological Systems theory, mixed-methods investigation were conducted for 61 Kalpar households with 245 residents via PRA tools along with HCD methods, that were triangulated through household interviews, brainstorming and focus group discussions. The results depicted that financially poorer families spend 2.36% of income on

water versus 1.06% for wealthier households. Agricultural abandonment forced 60% into fishing, 20% into Prosopis harvesting. Youth out-migration (20% male children, 10% adults) has caused demographic decline. Community-constructed problem trees emphasized social dimensions such as migration, debt, and depopulation alongside environmental factors, revealing lived experiences that were invisible to technical TDS measurements. SES analysis documented key feedback loops in the community. Coastal salinity is an intertwined socio-ecological system requiring integrated interventions. The PRA-HCD-based approach revealed community solutions that could supplement the persistently failing infrastructure-focused government programs. The proposed framework offers a replicable participatory climate adaptation model, challenging 'water-as-engineering-problem' paradigms and demonstrating the necessity of community epistemologies in sustainable development.

Keywords: participatory rural appraisal, sustainable, water salinity, socio-ecological systems, desalination

1. INTRODUCTION

Availability of safe drinking water, a key focus of the United Nations Sustainable Development Goals (SDG 6), is a basic need that a large proportion of the world population is still deprived of. Saline or contaminated drinking water is a stark, disturbing reality that millions of world population wake up to daily. Globally, 97.5 percent of (primarily seawater in oceans, with smaller contributions from saline groundwater and saline closed lakes) water is saline, and in India the tally marks around 95 percent. For families as well as farmers who depend on such water sources, the availability and sustainable management of clean water for a healthy living as well as for means of livelihood poses to be a crisis.

One of the standard metrics to measure water quality is the Total Dissolved Solids (TDS), which measures minerals, salts, metals, and ions in water, and is reported in milligrams per liter (mg/L) or parts per million (ppm) (Chidambaram et al. 2011). The World Health Organization (WHO) and Bureau of Indian Standards (BIS) advise a preferable TDS threshold of 300 mg/L along with a maximum limit of 500 mg/L for drinking water in India, and irrigation water should not go above 1920 mg/L. When TDS levels are beyond the threshold, people suffer with stomach aches, kidney problems, heart issues, nutrient imbalances, dehydration risks and bitter-tasting

water. For farmers, high TDS causes the crops to fail and the soil to degrade. This, in turn, threatens their income and food supply. Clean drinking water, beyond the livelihood, is about the basic dignity of families to stay healthy and communities working to sustain their way of life. In India, this water crisis affects 120,000–150,000 villages, home to 250 million people who rely on groundwater with TDS above 1000 mg/L (Raja and Neelakantan 2023). Desalination could help, but it is costly and harms the environment. A hydrogeochemical study undertaken in Ramanathapuram district of Tamil Nadu portrays the challenge as only 10 percent of groundwater samples are observed to be of good quality, while 10 percent were found to be marginally saline. 4 percent samples were reported to be saline, 1 percent marginally alkaline, 10 percent alkaline, 46 percent high in sodium adsorption ratio (SAR) saline and 19 percent highly alkaline (Karthikeyan et al. 2022). The statistics clearly reflect the gaping reality that a large section of the population cannot depend on their well water for consumption or irrigation (Ramesh et al. 2022).

The study entailed herein focuses on Kalpar, a coastal rural village in Ramanathapuram where the water crisis shapes daily life. In Kalpar, salty water causes scarce drinking water, failing crops, health problems, and environmental damage. This study undertakes participatory rural appraisal (PRA) techniques (N Narayanasamy 2009) as well as human-centered design (HCD) concepts (IDEO.org 2015) to listen to the local communities in Kalpar for getting a realistic appraisal about the woes of saline water as well as its impact on their daily lives. The interactive study strives to be a pilot project for investigating about cause-effect relationship between various socio, economic as well as environmental factors and to infer sustainable solutions for effective mitigation of the crisis.

1.1. Motivation

Coastal communities worldwide face existential threats from groundwater salinity – a crisis exacerbated by climate change and unsustainable practices. Despite global commitments to SDG 6 (Clean Water) and SDG 13 (Climate Action), most policy implementations often overlook hyper-local realities. In Tamil Nadu, where 68 percent of rural households rely on saline-prone aquifers, top-down interventions have failed to address disparities in water access (M. Baskar et al. 2022). This gap between policy intent and grassroots viability demands a paradigm shift. Community-centric design integrated with hydrological science has become a need of the hour to build equitable resilience. While hydrogeological studies extensively map salinity intrusion, few engage

with the socio-political dimensions shaping water use. The existing literature critiques the rift between the policy and practice but offers limited methodologies to remedy it. Crucially, the role of Participatory Rural Appraisal (PRA) and Human-centered Design (HCD) in co-creating culturally adaptive solutions remains underexplored – especially in contexts where traditional knowledge contradicts engineered fixes. This study responds to the aforesaid lacunae through a mixed-methods investigation of Kalpar - a rural coastal village in Ramanathapuram district Tamil Nadu, India, where salinity renders 100 percent of wells non-potable. A problem tree analysis revealed causal chains and cascading impacts. This systems-level view exposes flaws in current policy, which prioritizes infrastructure expansion over behavioral or governance reforms. By centering community voices in solution design, the dominant 'water-as-engineering-problem' narrative is challenged and the study advocates for embedded resilience.

1.2. Contribution

The following case study

- explores a framework merging PRA-derived causal analysis with HCD prototyping and SES framework, validated through hydrogeological surveys. The proposed approach quantifies participatory data, with well water salinity levels, revealing behavioural trends in the local community that are otherwise invisible to satellite mapping alone.
- identifies key areas for augmenting the governmental water policy and realigning the policies with the local realities
- proposes co-designing of sustainable intervention such as subsidized desalination plants mindful of community-driven, sustainable, cost-effective maintenance capacity; rainwater harvesting schemes that address the caste-based water-sharing norms; and soil fertility augmenting fuelwood trees as alternatives for invasive crop species. The interventions offer a blueprint for polycentric governance in SDG implementation.

1.3. Research Questions

Despite extensive hydrogeological documentation of coastal salinity in Tamil Nadu, three critical knowledge gaps persist: (1) the cascading socio-economic impacts of salinity remain unquantified at household level, (2) the value of integrating participatory methods (PRA) with design thinking (HCD) for co-creating solutions is underexplored, and (3) feedback loops between environmental

degradation and human responses are conceptually acknowledged but empirically undocumented in Indian coastal contexts.

This study aims to address the aforementioned gaps through three research questions grounded in Socio-Ecological Systems (SES) theory (Ostrom 2009)

1. What are the household-level economic burdens and livelihood impacts of groundwater salinity in Kalpar, and how do these impacts cascade to community-scale demographic changes?
2. How does integrating PRA with HCD reveal community-prioritized problems and causal mechanisms that differ from technical problem framings?
3. What feedback loops connect environmental degradation (salinity, invasive species) with social responses (livelihood choices, migration) in Kalpar's socio-ecological system?

2. LITERATURE REVIEW

Ramanathapuram district located in the south-eastern coastal edge of Tamil Nadu state in India faces a persistent water shortage due to its semi-arid climate, low rainfall, and saline groundwater due to climatic changes as well as sea intrusion. An average quantity of 827 mm annually rainfall is reported to reach the district, which does not fare well in comparison to the state-wide average rainfall of 925 mm. Owing to the poor quality of the local groundwater, as well as the comparatively higher needs for the penetration as well as grassroot delivery of the public supply initiatives such as the Jal Jeevan Mission (JJM), people in Ramanathapuram depend much on purchased drinking water. Groundwater salinity is another significant problem that plagues the coastal communities, which arises due to the entry of seawater and inappropriate geology. The geology beneath the surface of the area, which comprises coastal alluvium, clay, and sandy clay, fails to hinder the entry of saltwater into the groundwater system. Groundwater hydro-chemical characterization studies show higher values of sodium, chloride, and total dissolved solids (TDS) – usually higher than WHO and BIS recommended limits for safe drinking purposes (Raja and Neelakantan 2023). For instance, the presence of Chloride crosses 250 mg/L. The presence of other impurities such as sulphate, nitrate, and microbial contaminants makes the sources of the waters unfit for drinking purposes as well (S et al. 2025).

An ecological cause of the crisis is the spread of the alien invasive tree species Mesquite (*Prosopis juliflora*), known locally as Karuvelam. The tree has spread rapidly through the dry regions of Ramanathapuram (Chidambaram et al. 2011). Previous research conducted has shown that P. juliflora reduces the water tables of drylands considerably, modifying native ecosystems and competing with native vegetation (Dzikiti et al. 2013). As a part of the initiative to improve access to safe drinking water, the Government of India initiated the Jal Jeevan Mission (JJM) during 2019, targeting 55 liters per capita per day of safe drinking water through functional household tap connections (FHTCs). In Ramanathapuram, the process of implementation of the JJM includes combined drinking water supply schemes tapping the Cauvery river, along with other outside sources. As per our first-hand experience during appraisal, the evidence suggests that the supply of water continues to be irregular – in the range of once a week to once a month. The weak infrastructure of treatment, lack of quality testing, and lack of recharge mechanisms for water are reasons responsible for the slippage, through which initially workable schemes fail to maintain their sustainability over the long term. The erstwhile Ramanathapuram Mega Drinking Water Supply Project, inaugurated in 2007, continues to persist despite source depletion as well as delivery inconsistency. Traditional infrastructure such as water tanks, which used to be used for temporary summer storage of collected waters, have been reinitiated as part of the Rajiv Gandhi National Drinking Water Mission. Most such water bodies, nevertheless, remain silted, poorly maintained, and contaminated, and are disease breeding centers during the monsoons (S et al. 2025).

For above stated reasons, the use of above sources as potable supplies remains constrained, particularly for risk groups such as children and the elderly. UN SDG 6 aims at universal coverage of clean water and sanitation facilities by 2030, but coastal saline water in villages like Ramanathapuram district hinders progress, with a greater prevalence of chlorides at 250 mg/L or higher (S et al. 2025). It impacts health, causing hypertension, and limits agriculture, reducing productivity by 20% or higher. Hygienic water is central to good health as well as sustainable livelihoods, supporting 60% of rural workers in India. Participatory Rural Appraisal-Human Centered Design (PRA-HCD) (Ramesh et al. 2022) is a scientifically justifiable approach towards co-designing solutions, that is used to co-design water systems in Indian villages.

Integrated HCD-PRA framework implemented in two Indian villages with application of tools of PRA and principles of HCD in co-design of sustainable solutions towards water scarcity among others (S et al. 2025). It reports a successful case of PRA-HCD with 85 percent uptake among the community due to participatory action. Apart from environmental and infrastructure issues, water crisis is further exacerbated by socio-economic issues. Limited literacy, weak awareness of governmental initiatives, and low skill-building opportunities constrain adaptation capacity. The directives by the JJM prescribe IEC campaigns to have wider participation, though such initiatives have not made a noticeable difference. Water infrastructure like pipelines and RO plants are unused due to weak community ownership and low maintenance training. In brief, the drinking water condition of Ramanathapuram is caused by a multi-dimensional interplay of saline groundwater, alien species, suboptimal infrastructure, and deep-rooted poverty. Though policy actions like the JJM and desalination schemes are rich with promise, their success relies on sustained implementation, the engagement of people, and connection with long-term recharge and maintenance activities.

Socio-Ecological Systems (SES) framework (Ostrom 2009; Shi and Ling 2025) theory propounds that environmental problems cannot be understood in isolation from the social systems that depend on and shape them. In Kalpar, the ecological subsystem (groundwater, soil, and invasive species) interacts with the social subsystem (livelihood choices, migration decisions, community governance) through resource use patterns and feedback loops. This study pursues SES framework to analyse the water salinity crisis in Kalpar.

3. METHODOLOGY

This study adopts human-centered design (HCD) concepts (IDEO.org 2015) and synergizes it with participatory rural appraisal (PRA) tools (N Narayanasamy 2009) to investigate the day-to-day lives of rural coastal community. While the PRA tools enable the study to take part in the daily activities of community members and empower the community members as well as stakeholders to be vocal in representing their problems, the HCD is a broader, creative approach to problem-solving that places the needs of the community members at the core of the design process. The data from each source complement the other and enable the study to address the bias from either source through triangulation and cross-verification of data points. The study pursued an in-situ, ethnographic approach to understand the dynamics of rural life in Kalpar and to explore

various facets of the lives within the community (Ramesh et al. 2016). PRA is a participatory approach which systematically records local knowledge relating to marginalized rural communities. PRA employs many methods to interact and understand the community while at their moments of impediments within the settings where the pain points exist, thereby facilitating the appraisers to feel and comprehend the situation by participating and assessing on ground. Such methods ensure the broadening of the horizons for appraisers, building a connection with the community members, and understanding the intricacies of the actual problem as an insider or a stakeholder, rather than assuming problems as an outsider (Abhinaya et al. 2022).

A **resource map** is a visual representation that shows the location and distribution of key resources within a specific area. It is created in context to identify and analyze the availability of local resources, community problems, as well as to understand the village environment and infrastructure conditions of Kalpar. The **transect walk** enables the appraisers to explore and review the physical, topological as well social layout of the village. This tool assists in identifying those specific areas where the development is needed and the street where community is lacking the essential services. As a preliminary to transect walk, a *transect path* is identified. A transect path covers different areas that highlight problem mentioned and the main resources of the village such as ration, water tanks, primary health care center. **Household interviews** are conducted to gain insights into the living standards and conditions of the families in the community. Owing to the small size of population, the study interviewed all 61 households in the village without sampling. Focusing on *seven thematic areas*, a set of *structured* questions for initial day as well as *semi-structured* and *open-ended* interview questions for subsequent day are prepared to identify possible problems plaguing the community. HCD tools such as **AEIOU** and **six senses** are used to keenly note down critical aspects that govern the community during the transect walk as well as household interview. Interactions and participations with a set of planned questions are used to build the PRA tools. The interview responses are used to supplement the development of further PRA tools.

A **Venn diagram** is used to understand how various institutions, organizations, and people affect the community. The Venn diagram is made up of circles, each drawn for each entity (such as vendors, service providers, or organizations), the position of circles shows accessibility or proximity from the village, and the size of the circle shows how impactful a particular entity is to

our community. **Seasonal calendar** is made by interacting with the community members piecing together the temporal variation of socio-economic and cultural factors that influence the community. Seasonal calendar helps in acquiring a concise view of the cyclicity of changes that take place throughout the year pertaining to community such as customs, festivals, agrarian practices, health, weather, season, and climate. **Income-Expenditure** tool tabulates the financial resources of families and understand household-level income and spending patterns. This PRA tool helped us understand economic conditions and living standards of the community people. **Inflow-Outflow** tabulation helps identify the critical dependencies of the community by charting the resources that were exported from the village as well as the resources that were imported into the village.

The responses from individual households are triangulated, synthesized, and distilled to create core problems focusing on thematic areas, possible causes and results effects that impact the community, ruling out personal bias in individual responses. Further, each core problem is converted into a **problem tree**. A problem tree breaks down the complex problem into their root causes and visible effects. Problem trees help the researchers and stakeholders obtain a clear understanding of how a particular problem originates and evolves over time. The individual participant perceptions as well as assumptions about a core problem at hand represented as first draft of problem tree is further treated and sanitized with the collective intelligence of the crowd. **Brainstorming** is a meeting with the community leaders and community members to validate with a collaborative consensus, the individual participant assumptions, and perceptions about a core problem at hand that are identified during household visits. For each problem identified, a **focus group discussion** comprising of most affected parties and stakeholders is also devised. The sessions strive to identify and validate further causal reasons for the core problem as well as effects. Incorporating the suggestions and feedbacks from the brainstorming as well as focus group discussions, the final draft of problem tree is synthesized.

AEIOU framework serves a structured lens for ethnographic observations that reveals key insights into the community. The **six senses** engage the sight, sound, taste, smell, touch and intuition to deeply draw attention to experiences while with the community. The data acquired using PRA-HCD are further consolidated to develop **personas**. A persona is fictional yet hold key segments relating to the *pain points* of the community. Personas help capture the core problem by

reflecting their goals, challenges, pain points, and aspirations of the community members. A **scenario** mainly focuses on the pain points of the community thereby linking them to the problem tree. Scenarios define the struggles that the community members face in their everyday life and highlights the community needs for which meaningful inventions must be implemented.

This study employs the **Socio-Ecological Systems (SES) framework** (Ostrom 2009) to analyze water salinity as an integrated system where biophysical and social subsystems interact through feedback mechanisms rather than as isolated environmental or social problems. The SES approach conceptualizes crisis through four interacting components: resource systems (groundwater aquifers, coastal ecosystems), resource units (well water, agricultural land), governance systems (Jal Jeevan Mission implementation, Village Water Committees), and users (fishing households, Prosopis harvesters, migrating youth). SES maps causal chains identified through PRA problem trees onto feedback loop diagrams that reveal reinforcing cycles. The systems-level analysis exposes how interventions targeting single components such as installing RO plants fail when they ignore behavioral adaptations, governance capacity, and ecological feedbacks. The SES framework provides analytical scaffolding to integrate mixed-methods data (household economics, environmental surveys, community problem prioritization) into a coherent model of systemic vulnerability (Shi and Ling 2025).

4. RESULTS

Figure 1(a) depicts the resource map drafted using proposed methodology. The census data prepared before panchayath elections in 2019 indicated that the Kalpar had 70 houses with a population of 245 comprising 196 adults and 49 children. As per the household interviews, it was observed that 61 houses had residents. Of these, 20 percent were children while 78 percent were adults. 2 percent had been deceased between the census and household survey. 8 percent were male children while 12 percent were female.

4.1. Groundwater Salinity and Household Impact

As indicated by the resource map, only 37 percent of the 61 houses in Kalpar village had functional wells. In addition, there were 2 are public wells. Two-thirds of the community population is pushed towards buying costly reverse osmosis (RO-treated) water for all day-to-day needs. The Bay of Bengal is about 2 km south to the village Kalpar. The study reveals that all 23

households with wells have reported that their well water became qualitatively saltier over the past 20 years. Table 1 illustrates the results of water samples tested from approximately 75 percent the wells selected through random sampling. The unacceptable levels of TDS (<1200 mg/L acceptable) and hardness (200-500 mg/L acceptable) corroborate the insights shared by the community members. It was also identified that the salinity varies throughout the year and the same was triangulated using seasonal calendars and validated with community during brainstorming. The poor water quality has forced the entire community to depend on external water sources such as private vendors and tanker services for clean drinking water.

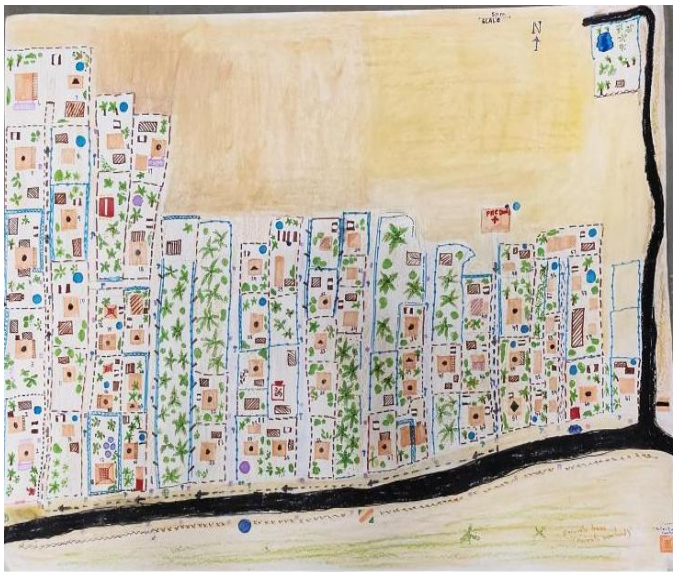

Fig. 1(a) Resources map of Kalpar village drawn with the community-driven initiatives

Fig 1(b). Legend for resource map of Kalpar village mentioned in Fig. 1 (a)

The Income-Expenditure mentioned in Figure 2 further indicated that the average household spending on clean water was ₹585, 2.5 percent of the average monthly household income of ₹23,217. However, the expenditure towards clean water represents 2.36 percent of total income for the poorest households (₹300 out of ₹12670), compared to 1.06 percent for wealthier families (₹480 out of ₹45000). In some cases, the expenditure towards water, equal to or greater than minimal food and energy consumption, indicate the unsustainable economic cost incurred due to salinity of water. This regressive financial distress exemplifies environmental injustice, where marginalized coastal communities subsidize their own water access while government infrastructure remains non-functional. The unequal infrastructure distribution reflects procedural

injustice (David Schlosberg 2007) due to historical underinvestment in marginalized coastal settlements. The SES framework reveals how the limited access to clean well water creates feedback loops: households without wells depend on costly vendors, reducing capital for livelihood diversification, perpetuating poverty, and vulnerability.

Table 1: TDS and Hardness of Water Samples from Kalpar

Sl. No	Sample Id	pH	Hardness mg/L	TDS mg/L
1	Kalpar-6	7.36	1100	3610
2	Kalpar-15	7.29	1805	5720
3	Kalpar-12	7.72	2020	9040
4	Kalpar-4	6.96	1940	6600
5	Kalpar-18	7.15	3100	9570
6	Kalpar-2	7.51	1540	5060
7	Kalpar-11	7.17	1160	3710
8	Kalpar-16	7.69	1000	5940
9	Kalpar-7	7.37	800	1790
10	Kalpar-1	7	1850	5870
11	Kalpar-5	7.57	480	2390
12	Kalpar-19	7.49	575	3020
13	Kalpar-6	7.36	1760	6730
14	Kalpar-10	7.21	1140	4400
15	Kalpar-13	7.44	575	3420
16	Kalpar-3	7.06	850	3240
17	Kalpar-14	7.15	1160	4480
18	Kalpar-8	6.84	2285	6270
19	Kalpar-9	7.27	1020	2650

House Number		1	2	3	4	5	6	7	8	9	10	Mean	Median
No of Members in Household		3	3	2	5	3	4	3	3	3	3	NA	3
Monthly Income per Household for the Current Year (₹)	Primary Employment	0000	20000	12000	15000	20000	45000	10500	25000	20000	15000	21250	20000
	Secondary Income	0	0	0	0	0	0	10000	0	0	8000	1800	0
	Other Income (Cattles)	750	0	0	0	0	0	0	0	250	0	100	0
	Govt Financial Assistance	0	0	670	0	0	0	0	0	0	0	67	0
	Total Household Income	**0750**	**20000**	**12670**	**15000**	**20000**	**45000**	**20500**	**25000**	**20250**	**23000**	**23217**	**20375**
	Transport	500	1200	100	2430	500	210	1500	500	500	1000	844	500
	TV Bills	230	230	230	230	230	230	230	230	230	230	230	230
	Mobile recharge	300	600	200	600	800	1300	750	600	500	200	585	600
	Ration	0	200	150	150	150	155	185	185	165	135	147.5	152.5
	Groceries	6000	3000	1200	1920	2550	1300	3440	2400	3800	1700	2731	2475
	Drinking Water	300	480	300	900	300	480	2250	300	240	300	585	300

Fig. 2: The income-expenditure for an average household in the community

4.2. Environmental Indicators and Agricultural Disruption

Approximately 55 percent of the land in Kalpar is owned by two individuals who reside outside the village while the 61 households own 45 percent. The dependence on seasonal income for livelihood led to vicious debt traps which forced villagers to sell off the agricultural land and migrate from the village. Based on PRA tools such as resource map shown in Figure 1 (a), it was deduced that 20 percent of the land in the village is occupied by settlements while 38 percent has some form of green cover. The remaining area is rendered arid and barren due to salinity and lack of water for irrigation. Of these 38 percent green cover, approximately two-thirds was covered by coconut and palm trees in the farmlands owned by the two wealthy outsiders. The land owned by the community members are predominantly covered with invasive *Prosopis juliflora.*

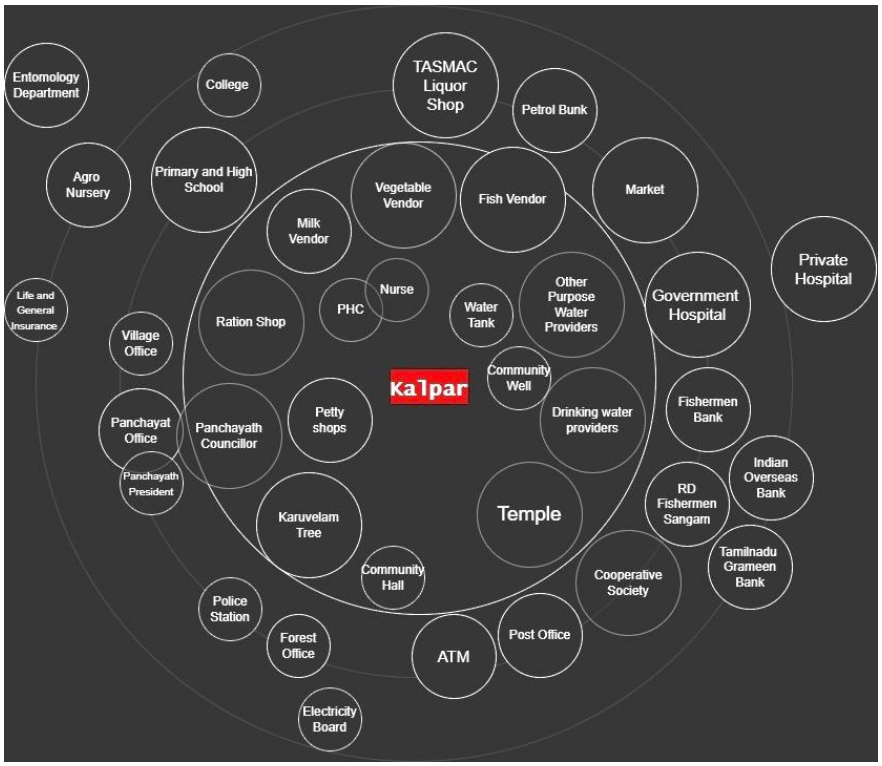

Fig. 3: Venn diagram depicting the stakeholders and key resources

The household interviews and brainstorming sessions outlined the community notion that only invasive species like *Prosopis juliflora* were found to be growing naturally in land owned by villagers. The high-water consumptive ability and competitive growth have caused them to be termed as problematic by the community. The environmental surveys by PRA tools such as resource maps and transect walks attested to the absence of agriculture in the whole village, even

with the availability of land. Transect walk indicated that palm trees, coconut trees, and tamarind trees grown sparsely in village have low economic use and do not contribute towards food or income security. Brainstorming consensus within the community suggested that *Prosopis juliflora* infestation that accentuated over the last 20 years was one of the predominant environmental factors toward worsening groundwater and soil conditions. One-fourth of the community support the invasive species as it contributes to the livelihood of woodcutting and produces economically tangible products such as fuelwood and charcoal.

4.3. Socio-Economic and Health Implications

The PRA tools revealed further implications of water salinity:

- The results of household interviews showed that health issues such as disease outbreaks were common in the community. Each household was affected by a water-borne disease at least once in the past year. The interviews reported multiple instances of presence of larvae in the stagnant and unused well water in addition to water stored in large containers over prolonged periods. The observations were reaffirmed by results of randomized checks from the epidemiology department of India since 2018-19 pandemic. The most persistent illness included diarrhoea, cholera, and typhoid with the women and children being the most vulnerable. Such illnesses not only pose health risks, increased financial stress and loss of attendance at school but put additional caregiving burden on women, limiting their ability to work or contribute to community development.
- The transect walk validated the community opinion that the village lacks an operational sub-primary health center within the village. As indicated in the Venn diagram depicted in Figure 3, the PHCs, private hospitals and schools are all outside the village, forcing people to travel for receiving basic services.
- Economic stress is further heightened by the unreliability of fishing (a seasonal income) and over-reliance on mesquite wood, which although economically rational, results in increased salinization and long-term environmental harm.

4.4. Community Stability and Migration

One of the most important effects of environmental deterioration is increased youth migration. The dropout rate among male children was seen to be around 60 percent. 33 percent

were adult female while 45 percent were adult male. Three-fifth of the population depended on the sea for livelihood. One-fifth of the population were dependent on woodcutting as a primary source of income. Less than 7 percent of the adults had a stable salaried source of income. Approximately 10 percent of the adults and 20 percent of the male children have migrated from the community looking for a stable source of income. The data was curated through household interviews and validated during brainstorming sessions. The distilled results were used to synthesize the scenario analysis and problem tree mapping. The lack of education and the conservative wisdom to be dependent on sea led to a high dropout rate among male children. The community members, especially youngsters, moved to neighboring towns or cities for alternative livelihoods, leading to depopulation, and undermining of the cultural identity of the village. Through community interviews and brainstorming, it was observed that there was a willingness to accept alternative livelihoods in case sustainable alternatives, particularly salt-tolerant agriculture, or micro-enterprise activities, were to be made available.

5. DISCUSSION

The study acknowledged the power asymmetries associated with extractive research paradigms when external experts study passive subjects in the local community. The study recognized the community members as knowledge holders about their own water crisis. The in-situ study performed iterative cycles of observation, reflection, and action. The results of Kalpar village provide a powerful demonstration of how environmental degradation – groundwater salinization in this instance – initiates a series of socio-economic effects in rural coastal communities. In contrast to blanket assertions of vulnerability, this research uses Participatory Rural Appraisal (PRA) and Human-centered Design (HCD) to produce statistically informed, locally driven evidence of the crisis (Ajith et al. 2022).

The consensus from all 61 households of higher water salinity, coupled with quantifiable economic costs of ₹7020 per year per household on clean water alone, decisively establishes the assertion of a water crisis being a quantifiable structural stressor. The lack of agriculture in the face of land holding and the prevalence of aggressive *Prosopis juliflora* – associated with groundwater drawdown – also justify environmental roots of the crisis (Shiferaw et al. 2024). These results corroborate larger studies on the effects of saline intrusion along the coasts of Tamil Nadu and highlight the ecological specificity of the issue. The research confirms that

environmental concerns like water salinity do not exist independently but are extensively intertwined with health, livelihoods, and demography. Families are economically strained, reliant on external water sellers, and suffer from water-borne diseases because of polluted well water. This economic burden creates trade-offs – between food, health, water, and the education of children – heightening poverty. In addition, farm decline, and the fishing seasonality limit income diversification, leading to youth migration. This, in its turn, leads to social cohesion erosion, as uncovered by persona-based scenario analyses.

The employment of the problem tree analysis and application of proposed SES framework was key to breaking down the core problem of salinity into its infrastructural, behavioral, and ecological causes. The perception of the community – accusing *Prosopis juliflora* – is supported by research and observed in the transect walk and resource map (Shiferaw et al. 2024).

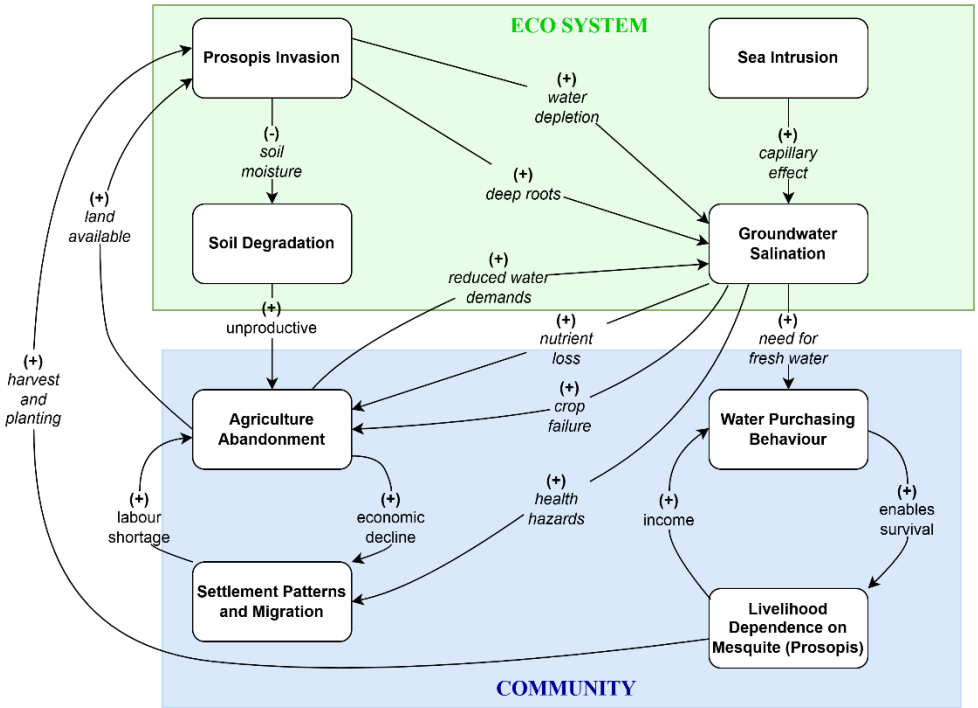

Fig. 4. Components and interactions in the reinforcing feedback loops of the mesquite trap and the salinity crisis

The illustration in Figure 4 models a socio-ecological system (SES) in the semi-arid coastal village of Kalpar. The system is characterized by a precarious balance between agricultural livelihoods, a rapidly transforming ecosystem due to an invasive species known as *Prosopis*

juliflora (mesquite), and the encroaching stress of salinity. The system portrays the tension between the short-term economic benefits of the invasive mesquite and its long-term ecological costs, which are exacerbated by and further exacerbate groundwater salinization.

Though the deep taproots of Prosopis deplete groundwater tables, the harvesting provides crucial short-term income in the form of labor or economic products for the community. The lower water tables and reduced freshwater recharge increase groundwater salinity by concentrating salts and potentially pulling in saline water. Increased salinity harms agriculture and domestic water supplies, increasing the need for purchased water. The revenue from mesquite enables households to afford this water, allowing the cycle to continue and even intensify. The system gets locked in a reinforcing cycle that progressively depletes and salinizes the groundwater base, increasing long-term vulnerability for short-term survival.

The increased salinity from above loop in addition to capillary effect from intruding sea leads to crop failure and agricultural abandonment, which in turn leads to economic decline at the household and community level. The economic decline and degraded environment triggers migration and scarcity of labor. The circumstances reduce the demand for irrigation water, which slightly helps slow down the increase of the groundwater salinization problem. The system achieves a balance from an ecological perspective as it reduces water demand, but at the cost of an undesired, unsustainable societal collapse through economic decline and out-migration.

5.4. Future Works

The study does not investigate governance failures in Jal Jeevan Mission implementation empirically, though policy implications are discussed based on literature. Future research should examine procedural justice dimensions through institutional ethnography and policy implementation analysis.

Lacking frequent monitoring of groundwater salinity, it is not possible to identify trends or assess interventions across time. Basic TDS meters, training initiative, and community-based water testing procedures could enable residents to monitor changes and respond accordingly (Dandadzi and Kothurkar 2023). Above all, technological solutions in themselves – desalination or RO systems – need to be complemented by behavioral training, financial planning for

sustainability, and Village Water Committee ownership for long-term sustainability (K et al. 2022).

6. CONCLUSION

This study investigates three critical gaps in coastal salinity research at Kalpar village, where 100% of wells are non-potable. First, the cascading household-level impacts are quantified. The results show that families spend ₹7,020 annually on water which is a regressive burden consuming 2.36% of income for the poorest versus 1.06% for wealthier households. The overdependence on sea for livelihood and conservative practices coupled with agricultural abandonment forced 60% of the adult population into fishing and 20% into Prosopis harvesting, triggering 30% out-migration of the youth population. Secondly, integrating PRA with HCD revealed community-prioritized problems such as debt cycles, depopulation, social fracturing that were otherwise invisible to technical TDS measurements (3,610-9,570 mg/L). The efficacy of participatory methods in exposing lived realities beyond engineering metrics is clearly demonstrated. Thirdly, two reinforcing socio-ecological feedback loops: the "mesquite trap" where Prosopis harvesting provides survival income while depleting groundwater further exacerbating salinity, and the salinity-migration spiral where environmental degradation drives out-migration, reducing labor for adaptive capacity is captured.

The proposed framework based on participatory appraisal triangulated with hydrogeological data, synthesized through problem trees and SES feedback modeling offers a replicable methodology for the 120,000-150,000 saline-affected Indian villages housing 250 million people. Sustainable climate adaptation emerges from systematically amplifying community epistemologies. Simultaneous intervention in ecological restoration, livelihood diversification, and procedural justice is essential for breaking vicious socio-ecological poverty traps. The study underscores that coastal salinity is an environmental justice crisis where marginalized communities subsidize their own survival; solutions must therefore prioritize equity alongside hydrology.

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